

Project Title: CampusSwap - Academic Resource Exchange System

Team Members

- Elif Doğa Tarhana - 2311504227
- Berkant Acun - 2111504040
- Uğur Pehlivan - 2311504262
- Melih Kaan Beşir - 2311504247
- Ahmet Emir Civelek – 2311504236

Overview

CampusSwap is a web-based system designed for university students to exchange academic resources. After logging in, users can create, update, and delete posts to share resources. To find materials, users can browse and search posts, and filter them by department. Students can also add posts they are interested in to their personal favorites for later consideration.

Students can send exchange requests to other users and manage these requests by accepting or rejecting them. During the exchange process, users can communicate via a dedicated messaging system tied directly to the trade. Once an exchange is successfully completed, users can rate and review each other based on their experience, helping to build a trusted community. Furthermore, to maintain a safe environment, users have the ability to report inappropriate behavior or posts. The main goal of this system is to make sharing educational resources easy, organized, secure, and accessible for all students.

Entities and Attributes

- **User:** Members of the platform have a unique user_id, username, email, password, and an avatar name. Each user is also associated with a specific department through a dept_id. The user_id is the key attribute. A user can have multiple skills at the same time (Multivalued).
- **Resource:** This entity represents the materials being shared. Each resource has a unique resource_id, a type (e.g., PDF, book), and a description. It is also categorized by a dept_id. The resource_id is the key attribute.

- **Post:** Users create posts to describe what they are offering and what they are requesting in return. A post includes a unique post_id, the user_id of the owner, an offer_id, a request_id, and the current status of the listing. The post_id is the key attribute.
- **Exchange:** This entity manages the trade requests. It tracks a unique exchange_id, the related post_id, the requester_id (the user asking for the trade), and the transaction status. The exchange_id is the key attribute.
- **Department:** To organize users and resources, the system stores department information. Each department has a unique dept_id, a name, and belongs to a specific faculty. The dept_id is the key attribute.
- **Review:** This entity allows users to rate their trade experience after an exchange is completed. It includes a unique review_id, the related exchange_id, the reviewer_id, the reviewee_id, a rating, and a comment. The review_id is the key attribute.
- **Exchange_Message (Weak Entity):** This entity stores the chat history between users negotiating a trade. It is a weak entity because a message cannot exist without an exchange. It has a partial key message_no and depends entirely on the exchange_id. Other attributes include sender_id, content, and timestamp.
- **Report:** This entity allows users to report other users or suspicious activities to maintain platform safety. It includes a unique report_id, the reporting user_id, a reason, and a status. The report_id is the key attribute.

Relationships

- A user can create multiple posts; but each post belongs to exactly one owner (1:N).
- A post links to specific resources; many posts can use the same resource from the system (N:1).
- Multiple users can request a trade for a single post; a post can have multiple exchange requests (1:N).
- A department can have many users; one user can have one department (1:N).
- A department can have many resources; a resource can have one department (1:N).
- A user can submit multiple reports; but each report belongs to exactly one user (1:N).
- A completed exchange can have multiple reviews (e.g., both the requester and the offerer can review each other) (1:N). A user can write multiple reviews and receive multiple reviews (1:N).
- **Favorites (N:M):** A user can favorite multiple posts, and a post can be favorited by multiple users. This relationship tracks the added_date.

- **Contains (Identifying Relationship):** An exchange request contains multiple messages. The Exchange_Message entity relies completely on the Exchange entity for its identification (1:N).

ER DIAGRAM

